MIL-B-1861A(DM)
2 June 1964

SUPERSEDING
MIL-B-1861
30 December 1949

MILITARY SPECIFICATION

BED, FOLDING, STEEL

1. SCOPE

1.1 This specification covers one type of steel folding hospital bed suitable for use in military field hospitals.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for.
TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-491	Enamel, Gloss, Synthetic (for Metal and Wood Hospital Furniture.
TT-P-659	Primer-Surfacer, Synthetic, Tints and White (for Metal and Wood Surfaces.
PPP-B-636	Box, Fiberboard.

STANDARDS

Federal

Fed. Std. No. 595 Colors.

FSC 6530

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 Outlet port sleeves or plugs not removable

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWINGS

Defense Medical Supply Center

2082 Bed, Folding
20740 Fiberboard Protector

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The material used in the construction of the beds shall be a good commercial quality, furniture grade steel.

3.1.1 Helical springs. Helical springs shall be formed from 0.1055 inch diameter (No. 12, Washburn and Moen gage) steel wire. Springs shall be heat treated after forming so as to comply with the test specified in 4.4.1.

3.2 Construction. The folding bed and component parts shall be in accordance with Drawing 2082. The bed shall consist of folding bed ends, bed frame, and spring fabric. The spring fabric may be of any type shown on the drawing. The bed shall be furnished complete with two mosquito-bar rods. The commercial tolerances for thickness of metal parts will be permitted. Other tolerances shall be as follows:

Operating parts of folding mechanism	± 1/64 inch
Other parts of folding mechanism	± 1/32 inch
Overall dimensions of tubular ends, spring frame assembly, and completed bedstead	± 1/16 inch

The bed shall pass the load test specified in 4.4.2.

3.3 Finish.

3.3.1 Plating. Plating of the assembled spring fabric shall conform to the requirements of Specification QQ-Z-325 for type I, class 3.

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3.3.2 Paint. All parts of the bed shall have the following painted finish.

3.3.2.1 Cleaning and preparation. Cleaning and preparation shall be accomplished as specified for grade I of Specification TT-C-490.

3.3.2.2 Primer. After cleaning and phosphate treatment, the parts shall be given one coat of a synthetic primer conforming to Specification TT-P-659. The primer shall be baked on for not less than 45 minutes at a temperature of 250° Fahrenheit (F.), or for an equivalent baking period at a higher or lower temperature, as recommended by the paint supplier and approved by the contracting officer. The primer shall be applied so as to level out and produce a uniform surface free from pebbles or wrinkles.

3.3.2.3 Enamel. After the primer has been allowed to cool, all surfaces shall then be covered with two coats of synthetic gloss enamel conforming to class B of Specification TT-E-491. Each coat shall then be baked on separately for not less than 45 minutes at a temperature of 250° F., or for an equivalent baking period at a higher or lower temperature, as recommended by the paint supplier and approved by the contracting officer. The enamel coats shall be applied so as to level out and produce a final, smooth, uniform surface without runs, wrinkles, orangepeel or grit, and without areas of thin or no film. The total dry-film thickness shall be not less than 0.003 inch at any point.

3.3.2.3.1 Color. Color shall be olive drab conforming to color X14087 (see 6.2) of Fed. Std. 595.

3.3.2.4 Touch-Up. After the spring fabric has been installed, each bedstead shall be touched up where necessary with a quick air drying olive drab enamel which shall conform in color and luster to the original baked-on olive drab film and which shall not deteriorate the baked-on film. The same touch-up enamel shall be used on the mosquito-bar assemblies where necessary.

3.4 Identification marking. Each bed shall bear a name plate, inconspicuously located, containing the following data:

Federal stock number.
Name of manufacturer or registered trade-mark.
Contract or purchase order number.
U. S.

Marking secured by adhesives shall not be used.

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upright position, these connections shall be at the top of the

3.5 Workmanship. The bed shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the levels set by the applicable quality levels.

3.5.1 Welding and riveting. The welding and riveting shall be durable and sound in every detail. The component parts of the bed shall be carefully and properly aligned prior to welding and riveting into permanent position, so as to provide a straight, rigid, and level assembly. All welds shall be accomplished in accordance with standard industrial practice. All ground welds shall be wire brushed smooth to eliminate traces of pittings and seams, and shall be polished to blend with adjacent surfaces. Fusion welds shall be accomplished by the heli-arc welding process using filler rod of appropriate composition.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105.

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red blood cells. The absence of

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total (major and minor combined) defects	II	4.0
For dimensional examination	S3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Standard MIL-STD-105. The sample unit of helical springs shall be one spring.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Helical springs	Permanent set	3.1.1	4.4.1	S1	*
Bed	Load	3.2	4.4.2	S2	*
Spring fabric	Plating	3.3.1	4.4.3	QQ-Z-325	

* Acceptance number shall be zero (0).

4.3 Examination. The beds shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major
101 Any component missing.
102 Any component fractured, split, cracked, punctured, cut, dented, sprung, malformed, creased, or otherwise impaired, affecting serviceability.
103 Any part misplaced, loose, or not in proper alignment, affecting serviceability.
104 Any operation of assembly omitted or not properly performed.
105 Any functioning component inoperative or will not operate as intended.
106 Any rivet missing, loose, or not properly set.

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107 Any weld missing or not complete.
108 Any weld burned through, cracked, or undercut.
109 Steel not free from seams, (except seams in welded tubing) laminations, blisters, excessive or loose scale or other defects which might affect appearance or serviceability.
110 Tubing bent to the radius shown in drawing not free from grooves, cracks or excessive roughness.
111 Malleable iron tees not strong, tough, even grain, sound, true to form, free from cold sheets, blow holes, porosity, brittleness or other defects.
112 Pivot points not accurately located; ends do not fold without binding or jamming.
113 Latches not precision formed.
114 Edges of legs of all angles not rounded.
115 Ends of braces, edges of latch cams and foot caps not free from burrs or fins.
116 Corner of spring frame assemblies, ends of end angles and surfaces of all angles at or adjacent to punched holes for springs, and all other parts not finished smooth, free from burrs, fins, cracks, fissures or rough projections.
117 Enamel coat not a smooth, uniform surface without runs, wrinkles, orangepeel or grit, and without areas of thin or no film.

Minor
201 Finish not as specified; not free of scratches, rough grind marks and other defects.
202 Unit not clean; not free of dirt, oil, and grease.
203 Any weld not free of flash, splatter, flux, burns, etc.
204 Any weld not ground and polished to blend with adjacent surfaces; not free of pitmarks and seams.
205 Identification marking missing, incomplete, illegible or not permanent.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The beds shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Helical spring. The length of the helical spring shall be accurately measured. Each spring shall support a 30 pound weight for 10 minutes without extending more than 1/2 inch. Each spring shall then be extended to 1-1/8 inches more than its nominal length for 10 minutes. After removal of the stress, the spring shall not have increased more than 1/32 inch in length.

4.4.2 Load. The bed complete with spring fabric shall be set up and ends latched in position. An initial load of 25 pounds shall be placed on the center of the fabric and the vertical distance measured from a mark on the center of the fabric to the floor, or to a fixed point independent of the bedstead. The initial load shall then be removed and a load consisting of 7 bags of cement (658 pounds), or sand in cement bags, weighing not less than 658 pounds, shall be placed on the spring as follows: A pile of three bags shall be placed at the center of the fabric with the long dimension of the bags transverse to the bedstead. Two piles of two bags each shall be similarly placed on either side of the center equidistant from the center pile and the end rails of the frame. Failure of any part of the bed shall be cause for rejection. After one hour the load shall be removed. The initial load shall then be again placed on the fabric as specified above, and the measurement repeated. The difference between the first measurement and the second measurement (permanent set) shall be not more than 1/4 inch. In addition, on completion of the test no helical spring shall have appreciable permanent set.

4.4.3 Plating. Tests of plated spring fabric shall be conducted as outlined in the applicable portions of Specification QQ-Z-325.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) bed, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	2 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit preparation. Each mosquito-bar rod shall be assembled in "T" form and the horizontal bars and vertical bar of each rod shall be covered with a kraft wrapping paper sleeve. The two assembled rods shall then be placed together and securely wired to the under-side of the fabric so they cannot move. The mosquito-bar guide on each tubular end shall be protected with an excelsior pad not less than 2-1/2 inches wide by 6 inches long which shall cover the guide and be securely tide to the tubing. The tubular ends shall then be folded under the frame.

5.2 Packing.

5.2.1 Level A. Each end of the frame of the bed shall be enclosed in a corrugated fiberboard protector as shown in Drawing 20740. The fiberboard shall conform to type I, class 2 of Specification PPP-B-636. A protector shall be slipped over each end of frame and fastened with not less than 4-ply twine laced through the holes in center edge of protector and tied in position. Two bedsteads shall be placed spring against spring and tightly secured together with three flat steel bands as follows: One band shall encircle the two bedsteads longitudinally, at approximate center. The remaining two bands shall be drawn around the two fiberboard protectors at each end of the bedsteads, transverse to the bedsteads, and over the longitudinal band. The bands shall be flat steel strap not less than 3/8 inch by .020 inch or 1/2 inch by .015 with a breaking strength of not less than 600 lbs. All bands shall be tightly drawn and sealed. The knot breaking strength of the steel bands shall be not less than 75% of the tensile strength. All bands shall have a rust resistant finish.

5.2.2 Level B. Beds shall be packed as specified in 5.2.1, except that fiberboard utilized shall conform to type I, class 2 of Specification PPP-B-636.

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5.3 Marking.

5.3.1 Exterior containers. Exterior containers shall be marked in accordance with Standard MIL-STD-129. The letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock Number" or "FSN" shall not be indicated.

6. NOTES.

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.2 Availability of color standards. The color standards of the color referenced in 3.3.2.3.1 are available from:

Commanding General
Aberdeen Proving Ground, Maryland
Attention: Coating and Chemical laboratory

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army.-MD
Navy.-BuMed
Air Force.-AFMSG

Preparing activity:
Defense Supply Agency.-IM

Review activities:
Army.-MD
Navy.-BuMed
Air Force.-AFMSG

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